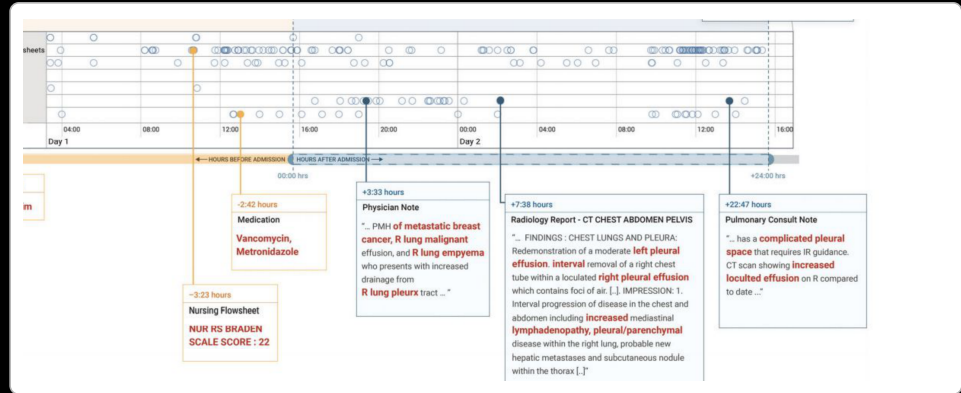


A patient is never one image

ONE ADMISSION, AND EVERYTHING WE RECORDED



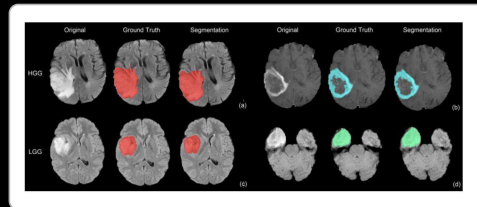
Two days of one admission: encounters, labs, orders, procedures, diagnoses, notes, medications. Rajkomar et al., *Scalable and accurate deep learning with electronic health records*, npj Digital Medicine

Every model in this course so far consumed **exactly one array**: an image in, a label or a mask or another image out. The chart above is what the hospital actually holds about this patient, and almost none of it is that array.

TWO FLAVOURS OF “MULTIMODAL”

Nearly identical

Four MR sequences of one brain. Same grid, same patient, same millimetre.



Barely comparable

A chest x-ray and the free text written about it.
No shared grid, no shared units, not even a shared data type.

Indication: Chest pain. Findings: mediastinal contours are within **normal** limits. Heart size is within **normal** limits. **No** focal consolidation, **pneumothorax** or **pleural effusion**. Impression: **No** acute cardiopulmonary abnormality.

Normal Report

```
def LP_pneumothorax(c):  
    if re.search(r'pneumo.*', c.report.text):  
        return "ABNORMAL"  
  
def LP_pleural_effusion(c):  
    if "pleural effusion" in c.report.text:  
        return "ABNORMAL"  
  
def LP_normal_report(c, thresh=2):  
    if len(NORMAL_TERMS.intersection(c.  
        report.words)) > thresh:  
        return "NORMAL"  
    LFs
```

The distance between the modalities is what decides the architecture, and that is the whole of the next section.

Left: Dong et al., MIUA. Right: after Khandwala et al.

FUSE ONLY WHAT IS COMPLEMENTARY

What each one knows

- ▶ The **image** localizes: where the lesion is, how big, what shape.
- ▶ The **report** names: which finding, how certain, compared to when.
- ▶ The **labs and vitals** give trajectory, which no single scan can show.

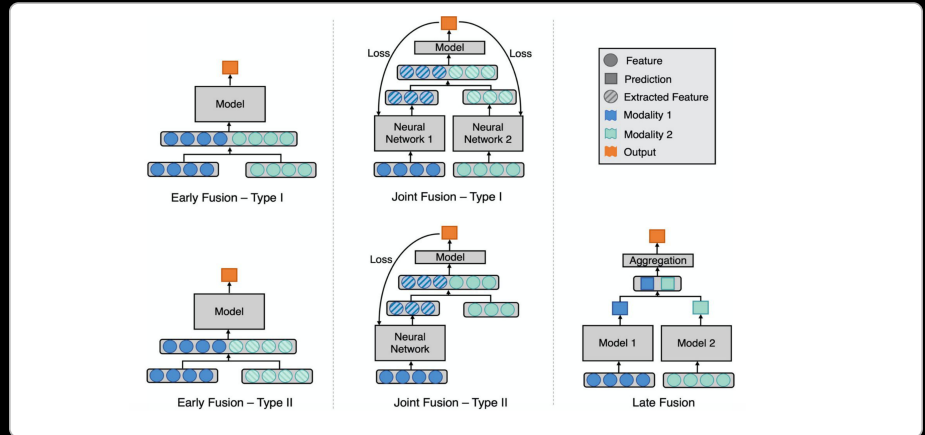
Before you add a modality

Ask whether it carries information the first one lacks. Two views of the same lesion are **complementary**; a second sequence that only restates the first is **redundant**, and it costs you parameters, acquisition time and one more thing that can be missing at test time.

A useful sanity check: train on each modality alone first. If the second one does not beat the first, fusion will not save it.

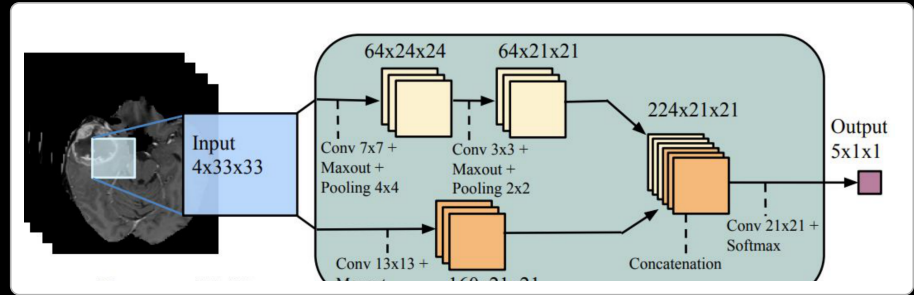
How to fuse

FUSION HAPPENS AT ONE OF THREE DEPTHS



- **Early:** combine the *data*, then run one model. Gradients never reach a modality-specific encoder.
- **Joint:** combine learned *features*; gradients flow back into every encoder. The common case.
- **Late:** combine *predictions*. Each modality is its own model.

EARLY FUSION: STACK THE MODALITIES AS CHANNELS



What changes

Nothing but the input depth. The brain tumour network takes $4 \times 33 \times 33$: one channel per MR sequence, exactly as an RGB image is three channels.

Why it is allowed

The sequences are **pixel aligned**, same resolution, same anatomy in the same place. The first convolution can mix them at a single voxel and the mixture means something.

WHEN STACKING STOPS WORKING

Different geometry

A CT volume and a mammogram view do not share a grid. There is no voxel at which to add them.

Different type

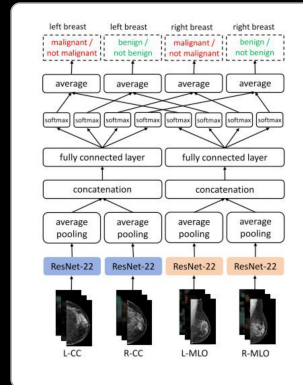
Text has no grid at all. Neither does a lab value, or a medication order with a timestamp.

Missing at test time

Stacked channels are mandatory. If one sequence was not acquired, the network has no input to run on.

The fix is the same in every case: give each modality its **own encoder**, and fuse further downstream, where both have been turned into feature vectors that live in comparable spaces.

JOINT FUSION: ONE ENCODER PER MODALITY, THEN CONCATENATE



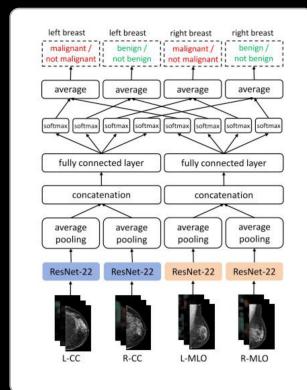
Breast cancer screening, four views

Four mammogram views (L-CC, R-CC, L-MLO, R-MLO), treated as four modalities. Each gets its own **ResNet-22**, then average pooling, then the four feature vectors are **concatenated** into one and a fully connected head predicts malignant and benign for each breast.

Why fully connected after the join

Concatenating feature vectors destroys spatial layout, so convolution has nothing left to slide over. Everything after the join is dense.

TWO DECISIONS HIDING INSIDE THAT DIAGRAM



1. Share weights, or not

L-CC and R-CC are the same view of different breasts, so they **share** an encoder. CC and MLO are genuinely different projections, so they get **separate** parameters. Similarity decides sharing.

2. Fuse once, or twice

This network does both: joint fusion by **concatenation** in the middle, and late fusion by **averaging** the CC and MLO predictions at the top. Mixing fusion types in one model is normal.

LATE FUSION: AVERAGE THE PREDICTIONS

What it is

Train a model per modality, then aggregate the outputs: average, weighted average, or a small model over the predictions. The modalities never meet until the very end.

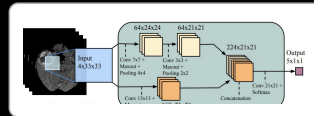
Why it is attractive in a hospital

Each model trains, validates and **fails** independently. A missing modality degrades the ensemble instead of breaking the input, and you can deploy the first model before the second exists.

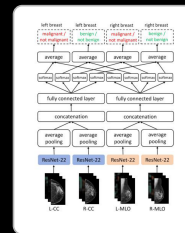
The cost: the modalities cannot help each other learn. Nothing in the image encoder ever sees the report, so no feature is shaped by the other modality. You get an ensemble, not a joint representation.

ASK THE ROOM: WHICH FUSION IS EACH OF THESE?

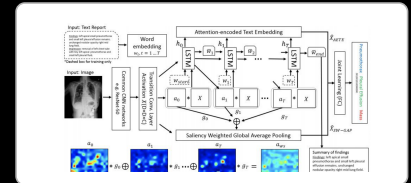
Havaei: four MR sequences



Wu: four mammogram views



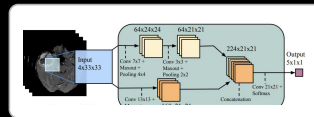
Wang: x-ray and its report



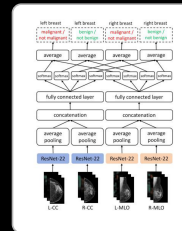
Early, joint or late? Look at where the arrows first meet, and at whether a gradient could reach both encoders.

WHICH FUSION IS EACH OF THESE?

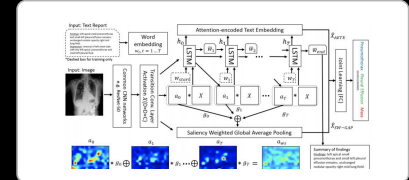
Early



Joint, then late

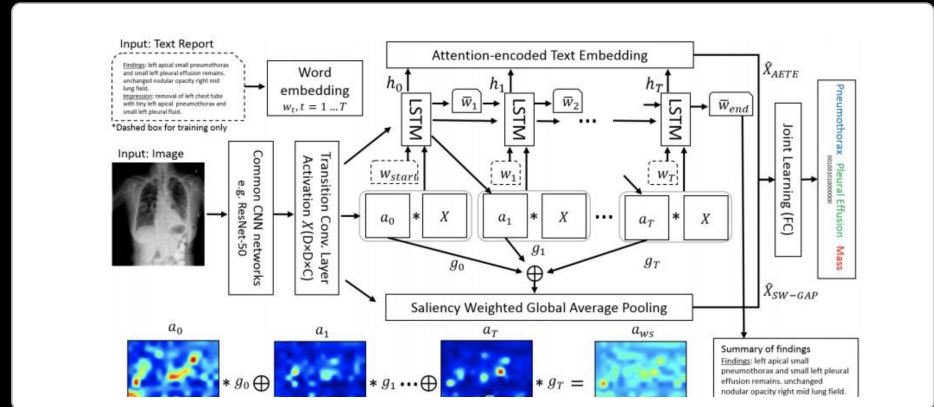


Joint



Havaei fuses before any learned layer. Wu concatenates features *and* averages predictions. Wang fuses learned image and text features inside the network. **Real models mix the categories:** the taxonomy is vocabulary for describing a design, not a set of boxes papers fall into.

IMAGE AND TEXT, FUSED WITH ATTENTION



A CNN encodes the x-ray; the report runs through a recurrent text encoder; the image features enter that encoder **at every step**, weighted by **attention**, so the model learns which region matters for which word. The fused representation predicts disease labels, which are then reused as **automatic annotations** for images that have a report but no label.